

Drift Assignment — How to Play

A one-page reference to every on-screen control and every setting.

Getting into the game

Install the APK on an Android device (min Android 7.0 / API 24, ARM64) and launch. You'll spawn at the start of the desert circuit, chase camera behind the car, engine idling. Everything below is playable from the first frame — no menu clicks required.

Editor testing? Use the **keyboard fallback**: `W` throttle · `S` brake · `A / D` steer · `Space` handbrake · `Q / E` shift down / up (manual).

Touch HUD — on-screen controls

The HUD is anchored to safe-area edges and mirrors a real driving cockpit. Nothing polls in `Update()` — every widget only redraws when its value changes.

Bottom-left cluster — steering + handbrake

Widget	What it does
← Left arrow	Steers left. Drag distance across the button becomes analog steering angle — a soft press turns gently, a full press turns hard. Release to auto-center.
→ Right arrow	Same behaviour, opposite direction.
Handbrake (bottom-center)	Press and hold to cut rear-wheel drive torque and drop rear-tire stiffness — the classic drift-initiation input. Release to grip up again.

Bottom-right cluster — pedals + gears

Widget	What it does
Gas pedal	Press and hold to apply throttle. Higher throttle spins the engine to the RPM redline (visible on the RPM readout).
Brake pedal	Press and hold to brake. If the car is at a full stop and you keep braking, the gearbox auto-shifts into Reverse .
Gear + / - buttons	Only interactive in manual transmission — dimmed and non-clickable in auto. <code>+</code> shifts up, <code>-</code> shifts down. In manual you can shift OUT of Reverse with <code>+</code> .

Top-right cluster — camera + settings

Widget	What it does
Camera button (icon swap)	Tap to cycle through 3 cameras: Chase (default follow) → Cinematic (orbit + look-ahead) → Broadcast (wide sideline). The icon and label update to show the current one. Hold the button to trigger the rear-facing look-back cam; release to snap back.
Settings ⚙	Opens the settings panel (see below). Game pauses (<code>Time.timeScale = 0</code>) while it's open.

Center readouts

Widget	Meaning
GEAR	R reverse · N neutral · 1 – 6 forward.
AUTO / MANUAL	Small label next to the gear text showing current transmission mode.
SPEED	Live km/h from the <code>Rigidbody</code> .
RPM	Engine speed. Updates via a smoothed value driven by wheel RPM × gear ratio.

Settings panel — 6 tabs

Tap the ⚙ icon to open. Every slider / toggle takes effect **immediately** — no "Apply" button needed. Tap the **backdrop** or the ✕ to close.

1 • Transmission

Control	What it does
Automatic ↔ Manual toggle	Switches shifting logic. In Auto , the gearbox picks the shift point from throttle + RPM. In Manual , the ± HUD buttons become active and shifting is entirely up to you.

2 • Suspension

Control	What it does
Front spring force	Higher = stiffer front, less nose-dive under braking, more understeer.
Rear spring force	Higher = stiffer rear, more traction on power exits but harder to initiate drift.
Front / rear ride-height	Lifts or lowers the chassis on that axle — affects centre of mass.

3 • Camber

Control	What it does
Front camber angle	Tilts the tops of the front wheels in / out. Aggressive negative camber increases cornering grip at the cost of straight-line tire wear.
Rear camber angle	Same on the rear axle — a common drift setup is mild negative rear camber for a controllable slide.

4 • Audio

Control	What it does
Master volume	Global gain on <code>AudioListener</code> .
Engine / SFX / UI sub-mixers	Individual gain lines for the engine 4-source mixer, gameplay SFX (screech, impacts, wind), and UI clicks.

5 • Graphics

Control	What it does
Quality preset	Switches Unity's Quality Level (Low / Medium / High). Affects shadow distance, texture streaming budget, and LOD bias.
Anti-aliasing	Off / FXAA / SMAA on the URP camera. FXAA is the mobile-safe default.
Post-processing on/off	Toggles the URP volume (Bloom + Tonemap + Color Adjustments + Vignette + Film Grain). Motion Blur / DoF / Chromatic Aberration are always off for mobile GPU safety.
Shadow quality	Off / Hard-only (25 m, low res) / All (80 m, high res). Big lever on low-end phones.

6 • About

Quick credits + links to the reference material and the docs bundle. Tap any `<link>` to open in the system browser.

Tips to actually drift

1. Get up to ~60 km/h in 2nd or 3rd gear.
2. Point the nose into the corner.
3. **Tap the handbrake briefly** to break the rear tires loose.
4. Counter-steer (opposite direction from the turn) to hold the slide.
5. Feed throttle progressively — too little and you spin out, too much and you snap back straight.
6. **Try the presets in Settings → Suspension** — stiffer rear + softer front makes the car much more drift-happy.

Damage feedback is live: light bumps leave dents, harder hits detach panels (bumpers, doors, mirrors). The car keeps driving through anything short of total loss — this is a demo, not a permadeath sim.